

Mosquitto MQTT Broker — Reference & Setup Guide

For the Ramzor bridge system | Personal reference notes

1. What it is

Mosquitto is the local MQTT broker that `train_dashboard.py` publishes to and subscribes from. It runs as a Windows service on the same PC as the dashboard, listening on `localhost:1883`. It should start automatically on every boot — the steps below are for checking/troubleshooting, not routine startup.

2. Quick reference — checking and restarting

Check whether it's running

```
Get-Service mosquitto
```

If `Status` shows **Running**, nothing else is needed.

Start it (if stopped)

```
Start-Service mosquitto
```

Restart it (e.g. after editing its config file)

```
Restart-Service mosquitto
```

Confirm it's actually listening on the network

```
netstat -an | findstr 1883
```

Should show `0.0.0.0:1883 LISTENING`.

3. Testing it manually

Full paths are needed unless Mosquitto's folder was added to PATH (see §4 step 5).

Subscribe (watch messages arrive)

```
& "C:\Program Files\mosquitto\mosquitto_sub.exe" -h localhost -t ramzor/bridge2/events
```

Publish a test message

```
& "C:\Program Files\mosquitto\mosquitto_pub.exe" -h localhost -t ramzor/bridge2/events -m  
'{"device":"test","type":"traffic_light","state":"red","reason":1}'
```

In PowerShell, use **single quotes** around the JSON payload with plain double quotes inside — backslash-escaped quotes get mangled by PowerShell's own parsing.

4. Installing on a new computer

Follow these steps in order when setting up the Ramzor system on a different PC.

Step 1 — Install Mosquitto

- Download the Windows installer from `mosquitto.org` (official Eclipse Mosquitto site).
- Run it with default options — this installs it as a Windows service (starts automatically on boot).

Step 2 — Edit the config file to accept network connections

By default, Mosquitto only allows local (loopback) connections. Since the dashboard, the bridge listener, and manual testing all need this too, add two lines to the end of the config file.

Open (as Administrator) or edit via PowerShell:

```
C:\Program Files\mosquitto\mosquitto.conf
```

Add these two lines at the very end:

```
listener 1883
allow_anonymous true
```

Quick way from an admin PowerShell:

```
Add-Content -Path "C:\Program Files\mosquitto\mosquitto.conf" -Value "`nlistener 1883`nallow_anonymous true"
```

Why this step is essential: without it, the config file's own default behavior only allows connections from the local machine itself. Without `listener 1883`, nothing external (including test tools) can reach the broker even though it appears to be running.

Step 3 — Restart the service to apply the config change

```
Restart-Service mosquitto
```

Step 4 — Open the firewall port

Windows Defender Firewall blocks unsolicited inbound connections by default. Add a rule for port 1883:

1. Windows Defender Firewall with Advanced Security → Inbound Rules → New Rule...
2. Rule type: **Port**
3. Protocol: **TCP**, Specific local ports: `1883`
4. Action: **Allow the connection**
5. Profile: apply to all (Domain, Private, Public)
6. Name it something clear, e.g. "*Mosquitto MQTT (1883)*"

Step 5 — (Optional) Add Mosquitto to PATH

So `mosquitto_pub / mosquitto_sub` can be run by name instead of full path:

1. Search "Environment Variables" → Edit the system environment variables → Environment Variables...
2. Under System variables, select `Path` → Edit → New
3. Add: `C:\Program Files\mosquitto`
4. OK through all dialogs, then close and reopen any PowerShell window

Step 6 — Verify

```
Get-Service mosquitto
netstat -an | findstr 1883
```

Then run `train_dashboard.py` and confirm the MQTT status indicator turns green.

5. Quick checklist summary

Step	Command / Action
Install	Run the Mosquitto Windows installer (default options)
Configure	Append <code>listener 1883</code> and <code>allow_anonymous true</code> to <code>mosquitto.conf</code>
Restart	<code>Restart-Service mosquitto</code>
Firewall	Add inbound TCP rule for port 1883

Verify

Get-Service mosquitto + netstat -an | findstr 1883